CMSC 420: Skip Lists

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January 20, 2026

1 Problems **2**

1 Problems

1. Suppose the keys 30,10,80,50,20 are inserted into a randomized skip list in that order.

   We know each element exists in level 0. When determine whether each element extends to the next higher level we use a random number generator which outputs either 0, which means the element extends no further, or 1, which means the element extends another level and we must check again.

   Draw the resulting randomized skip list under the assumption that the random number generator outputs the following:

   1,1,1,1,0,1,0,1,1,0,0,1,0

   Solution:
   From the random numbers we observe that:

   - The 30 extends to levels 1,2,3,4.
   - The 10 extends to level 1.
   - The 80 extends to levels 1,2.
   - The 50 does not extend.
   - The 20 extends to level 1.

   The final picture is then:

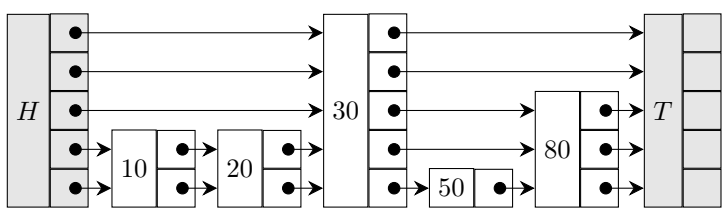

Figure 1: Resulting Skip List

2. Suppose a skip list has 100 nodes. What is the probability that at least one node will reach a level of 5 or beyond?

   Solution:
   The probability is:

$$1 - (1 - 1/2^5)^{100}$$

3. This problem digs deeper into the probabilities associated with a skip list with n nodes reaching a certain number of levels.

   (a) For each $k = 1, 2, 3, ...$ what is the probability that a skip list does not reach level k? Justify.

   Solution:
   The probability that one node reaches level 1 is $\frac{1}{2}$ and so the probability that one node does not reach level 1 is $1 - \frac{1}{2}$. Thus the probability that none of them reaches level 1 is $\left(1 - \frac{1}{2}\right)^n$.

   The probability that one node reaches level 2 is $\left(\frac{1}{2}\right)^2 = \frac{1}{2^2}$ and so the probability that one node does not reach level 2 is $1 - \frac{1}{2^2}$. Thus the probability that none of them reaches level 2 is $\left(1 - \frac{1}{2^2}\right)^n$.
   In general the probability that none of them one of them reaches level k, meaning the skip list does not, is:

   $$\left(1 - \frac{1}{2^k}\right)^n$$

   Note that this works for $k = 0$ too, since the probability that none of them reaches level $k = 0$ is $\left(1 - 1/2^0\right)^n = 0^n = 0$ since all the nodes reach that level!

   (b) For each $k = 0, 1, 2, 3, 4, ...$ what is the probability that a skip list does reach level greater than or equal to k? Justify.

   Solution:
   In general the probability that at least one of them reaches level greater than or equal to k is:

   $$1 - \left(1 - \frac{1}{2^k}\right)^n$$

   (c) For each $k = 0, 1, 2, 3, 4, ...$ what is the probability that the skip list reaches exactly level k? Justify.

   Solution:
   The probability that it reaches level greater than or equal to k but not greater than or equal to $k + 1$ is then:

   $$\left[1 - \left(1 - \frac{1}{2^k}\right)^n\right] - \left[1 - \left(1 - \frac{1}{2^{k+1}}\right)^n\right]$$

4. When building a skip list recall that there is a 0.5 probability that a node which reaches level i also reaches level $i+1$. Suppose we change this probability to some $0 < p < 1$.

   If $n = 80$ nodes are inserted into such a skip list what is the maximum value of p which ensures that the probability that at least one node reaches level 10 or beyond is 0.4 or below?

   Solution:
   The probability that at least one node reaches level 10 or beyond is:

$$P = 1 - \left(1 - p^{10}\right)^{80}$$

   To keep this below 0.4 we need:

$$\begin{aligned} 1 - \left(1 - p^{10}\right)^{80} &\leq 0.4 \\ \left(1 - p^{10}\right)^{80} &\geq 0.6 \\ 1 - p^{10} &\geq \sqrt[80]{0.6} \\ p^{10} &\leq 1 - \sqrt[80]{0.6} \\ p &\leq \sqrt[10]{1 - \sqrt[80]{0.6}} \end{aligned}$$

5. Suppose we build a skip list but we modify the probability such that each node has a p probability of extending to the next level.

   If the skip list has exactly one node, what do we expect the maximum level to be? Justify.

   Note: It may be useful to know that:

$$\sum_{i=0}^{\infty} ix^i = \frac{x}{(1-x)^2}$$

Solution:
The probability that the one node reaches exactly level L but no further is $p^L(1-p)$. It follows that the expected maximum level is:

$$\begin{aligned}\sum_{L=0}^{\infty} Lp^L(1-p) &= (1-p)\sum_{L=0}^{\infty} Lp^L \\ &= (1-p)\left(\frac{p}{(1-p)^2}\right) \\ &= \frac{p}{1-p}\end{aligned}$$

6. For a skip list containing n nodes and having enforced maximum level M, what is the probability that every node in the skip list reaches level M?

   Solution:
   The probability that a single node reaches level M (or higher, which truncates at level M) is $1/2^M$ so the probability that all of the nodes reach level M is $(1/2^M)^n$.

7. Suppose a skip list has n nodes and $p = 0.5$. What is the probability that the skip list reaches each of the following.

(a) Level 0 and Perhaps Further	1
(b) Level $k = 0, 1, 2, ...$ and Perhaps Further	$1 - \left(1 - 0.5^k\right)^n$
(c) Exactly Level $k = 0, 1, 2, ...$	$\left[1 - \left(1 - 0.5^k\right)^n\right] - \left[1 - \left(1 - 0.5^{k+1}\right)^n\right]$

8. Considering a single node in a skip list with probability $0 < p < 1$. For all $i \geq 0$, what is the probability it reaches level i but not futher?

   Solution:
   It would be:

$$p^i(1-p)$$

9. Suppose the keys 5,21,17,7,10,42 are inserted into a randomized skip list in that order.

   We know each element exists in level 0. When determine whether each element extends to the next higher level we use a random number generator which outputs either 0, which means the element extends no further, or 1, which means the element extends another level and we must check again.

   Draw the resulting randomized skip list under the assumption that the random number generator outputs the following:

   1,1,0,1,0,0,1,1,1,1,0,1,1,1,0,1,1,0,

   Solution:
   From the random numbers we observe that:

   - The 5 exists at levels 0,1,2.
   - The 21 exists at levels 0,1.
   - The 17 exists at level 0.
   - The 7 exists at levels 0,1,2,3,4.
   - The 10 exists at levels 0,1,2,3.
   - The 42 exists at levels 0,1,2.

   The final picture is then:

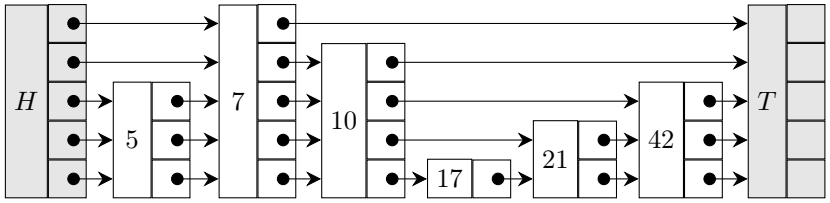

Figure 2: Resulting Skip List

10. We proved in class that if L is the maximum level in a skip list with n nodes then $E(L) = \mathcal{O}(\lg n)$. We proved the case where $p = 0.5$. Modify this for an unknown $0 < p < 1$. You may assume the definitions of I_i and n_i and you may assume that:

$$E(n_i) = np^i$$

Start at the point:

$$E(L) = \sum_{i=0}^{\infty} E(I_i)$$

Solution: We have:

$$\begin{aligned}
E(L) &= \sum_{i=0}^{\infty} E(I_i) \\
&= \sum_{i=0}^{\lfloor \log_{1/p} n \rfloor} E(I_i) + \sum_{i=\lfloor \log_{1/p} n \rfloor + 1}^{\infty} E(I_i) \\
&\leq \sum_{i=0}^{\lfloor \log_{1/p} n \rfloor} 1 + \sum_{i=\lfloor \log_{1/p} n \rfloor + 1}^{\infty} E(n_i) \\
&\leq 1 + \left\lfloor \log_{1/p} n \right\rfloor + \sum_{i=\lfloor \log_{1/p} n \rfloor + 1}^{\infty} np^i \\
&\leq 1 + \log_{1/p} n + np^{\lfloor \log_{1/p} n \rfloor + 1} \sum_{i=0}^{\infty} p^i
\end{aligned}$$

At this point note that:

$$\left\lfloor \log_{1/p} n \right\rfloor + 1 \geq \log_{1/p} n$$

And since $0 < p < 1$ we know:

$$p^{\lfloor \log_{1/p} n \rfloor + 1} \leq p^{\log_{1/p} n}$$

Thus, continuing:

$$\begin{aligned}
E(L) &\leq 1 + \log_{1/p} n + np^{\lfloor \log_{1/p} n \rfloor + 1} \sum_{i=0}^{\infty} p^i \\
&\leq 1 + \log_{1/p} n + np^{\log_{1/p} n} \sum_{i=0}^{\infty} p^i
\end{aligned}$$

Then observe that:

$$p^{\log_{1/p} n} = p^{\frac{\log_p n}{\log_p (1/p)}} = p^{\frac{\log_p n}{(-1)}} = \left(p^{\log_p n}\right)^{-1} = \frac{1}{n}$$

And so again continuing:

$$\begin{aligned} E(L) &\leq 1 + \log_{1/p} n + np^{\log_p (1/n)} \sum_{i=0}^{\infty} p^i \\ &\leq 1 + \log_{1/p} n + n(1/n) \sum_{i=0}^{\infty} p^i \\ &\leq 1 + \log_{1/p} n + \left(\frac{1}{1-p}\right) \end{aligned}$$

11. It can be shown that the expected maximum level in a skip list with n nodes and with $p = 0.5$ actually satisfies:

$$E(L) = \sum_{k=0}^{\infty} k \left[\left(1 - 0.5^{k+1}\right)^n - \left(1 - 0.5^k\right)^n \right]$$

This is very difficult sum to evaluate, so we won't. But however you like, evaluate or approximate this value for $n = 1000, 2000, 3000, ..., 10000$ to the nearest single digit after the decimal point.

Solution:
I tried using Wolfram Alpha but it often timed out so I did approximations via Python, basically summing the first ten million terms.

n	EML
1000	10.3
2000	11.3
3000	11.9
4000	12.3
5000	12.6
6000	12.9
7000	13.1
8000	13.3
9000	13.5
10000	13.7

12. Suppose a skip list with unknown $0 < p < 1$ has maximum enforced level $M \geq 0$. Let S_i be the number of steps going left in level i when following skip list search in reverse. We know that $E(S_i) \leq np^i$ for all $0 \leq i \leq M$. Use this single fact to provide a quick proof that:

$$E(S) = \mathcal{O}(n)$$

Solution:
We have:

$$E(S) = \sum_{i=0}^{M} E(S_i) \leq \sum_{i=0}^{M} np^i = n \sum_{i=0}^{M} p^i \leq n \sum_{i=0}^{\infty} p^i = n \left(\frac{1}{1-p} \right)$$

The result follows.

13. Suppose a skip list with $p = 0.5$ has maximum enforced level $M = \alpha \lg n$ where $\alpha \geq 0$. Let S_i be the number of steps going left in level i when following skip list search in reverse. Prove that the total number of steps S_L going left satisfies:

$$S_L = \mathcal{O}(\lg n + n^{1-\alpha})$$

Solution:
We have:

$$\begin{aligned}
\sum_{i=0}^{M} E(S_i) &= \left(\sum_{i=0}^{M-1} E(S_i)\right) + E(S_M) \\
&\leq \left(\sum_{i=0}^{M-1} 1\right) + E(S_M) \\
&\leq M + n(0.5)^M \\
&\leq \alpha \lg n + n(0.5)^{\alpha \lg n} \\
&\leq \alpha \lg n + \frac{n}{2^{\alpha \lg n}} \\
&\leq \alpha \lg n + \frac{n}{n^{\alpha}} \\
&\leq \alpha \lg n + n^{1-\alpha}
\end{aligned}$$

The result follows.

14. We proved that if S_i is the number of steps going left in level i when following skip list search in reverse satisfies $E(S_i) \leq 1$ when $p = 0.5$. This is for all i when no maximum enforced level is set. Construct a version of this for an unknown p. You may need to look up an unfamiliar series for this.

Solution:
If we are at the rightmost position on level i then we have:

- There is a probability of p that we go up, hence 0 steps left.
- There is a probability of $(1-p)p$ that we go left then up, hence 1 step left.
- There is a probability of $(1-p)^2 p$ that we go left twice then up, hence 2 steps left.
- And so on.

In general there is a probability of $(1-p)^i p$ that we go i steps left. Since we certainly don't go infinitely steps left we knwo that expected number of steps left satisfies:

$$\begin{aligned}
E(S_i) &\leq \sum_{i=0}^{\infty} i(1-p)^i p \\
&\leq p \sum_{i=0}^{\infty} i(1-p)^i \\
&\leq p \left(\frac{1-p}{(1-(1-p))^2} \right) \\
&\leq \frac{p(1-p)}{p^2} \\
&\leq \frac{1-p}{p}
\end{aligned}$$

Note that this isn't ≤ 1, that's specific to the $p = 0.5$ case.

15. Consider the following skip list:

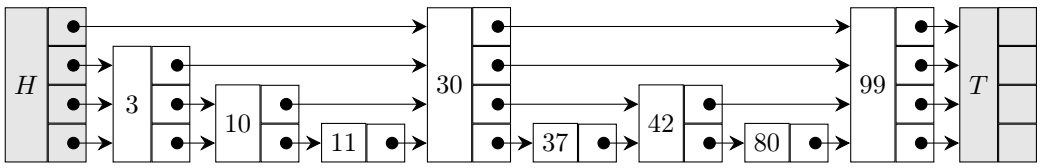

Figure 3: Skip List for the Problem

We know that search starts at the top of the header node and a *step* is either going down one pointer level or following a pointer. So for example it takes 5 steps to find 37 since we go F, D, D, D, F where F means follow a pointer and D means go down one pointer level.

(a) How many steps does it take to find 42?

How Many?	4

(b) If a key is chosen uniformly at random what is the expected number of steps it will take to find it? Show your work and simplify.

Solution:
We have:

$$\frac{1}{8}(2+4+6+1+5+4+6+2) = \frac{1}{8}(30) = \frac{15}{4}$$

16. Suppose a skip list has $p = 0.5$ and n nodes. No maximum enforced level.

(a) Suppose $n = 10$. What is the probability that at least one node reaches level 3 or higher? You do not need to simplify.

Probability	$1 - (1 - 0.5^3)^{10}$

(b) What is the minimum number of nodes that must be inserted so that the probability that at least one node reaches level 2 or higher is at least 0.7? No maximum enforced level. Simplify as much as possible; you can leave a logarithm in your answer.

Minimum Number	$\lceil \log_{0.75} 0.3 \rceil$

Scratch Work:
Both of these use the formula $prob = 1 - (1 - 0.5^i)^n$.

i. Just plug in $i = 3$ and $n = 10$.

ii. We require $1 - (1 - 0.5^2)^n \geq 0.7$ and so $0.75^n \geq 0.3$ and so $n \geq \log_{0.75} 0.3$. Since we need an integer we take the ceiling.

17. Suppose a skip list has $p = 0.5$ and n nodes. No maximum enforced level. Calculate the probability in terms of n (unknown) that every node has exactly the same height. Show your work and simplify your answer to get rid of any Σ sums.

Solution:
For some i, the probability that one node reaches exactly level i is 0.5^{i+1}. Since the nodes grow independently the probability that all do is $(0.5^{i+1})^n$. Then the probability taken over all i, since they are disjoint, is:

$$\begin{aligned} P &= \sum_{i=0}^{\infty} (0.5^{i+1})^n \\ &= 0.5^n \sum_{i=0}^{\infty} (0.5^n)^i \\ &= 0.5^n \left(\frac{1}{1 - 0.5^n} \right) \\ &= \frac{0.5^n}{1 - 0.5^n} \end{aligned}$$